A.2. MULTIPLE STAGE CONTINUOUS PROCESSES

A single partitioning of a compound between an organic solvent and water may not be sufficient for isolating it in acceptably pure form and good yield. Various multiple extraction techniques may therefore be required. Such techniques have been described in Section 21.6.4 and their technical application for uranium production (Section 14.2.4.3) and spent fuel reprocessing (Section 21.6.3).

Continuous processes are preferred in industry, where the most common and simple solvent extraction equipment is the mixer-settler (Fig. A.1(a); cf also Fig. 14.2). This type of equipment is also becoming standard in laboratories engaged in process development. In the uranium industry a single mixer-settler may hold as much as 1000 m^3. The mixer-settlers, each closely corresponding to a single ideal extraction stage, are arranged in

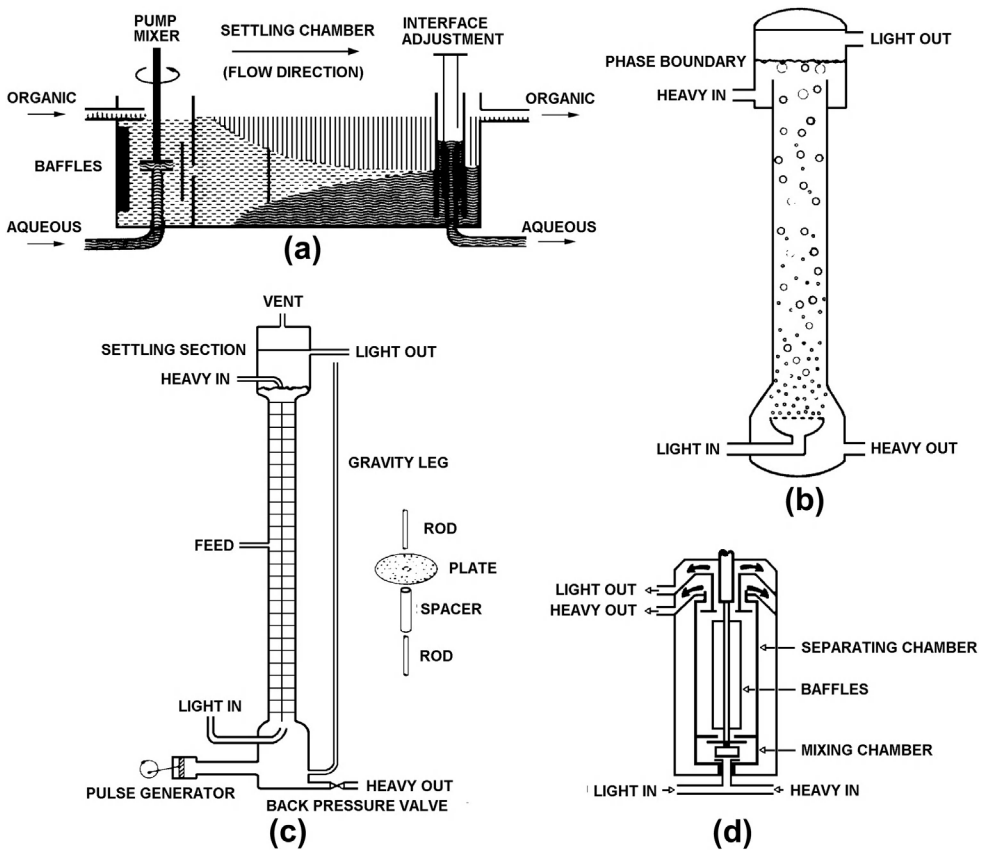


Figure A.1 Different types of continuous extraction equipment. (a) Mixer-settler (b) Spray column (c) Pulsed column (d) Tubular centrifugal contactor.

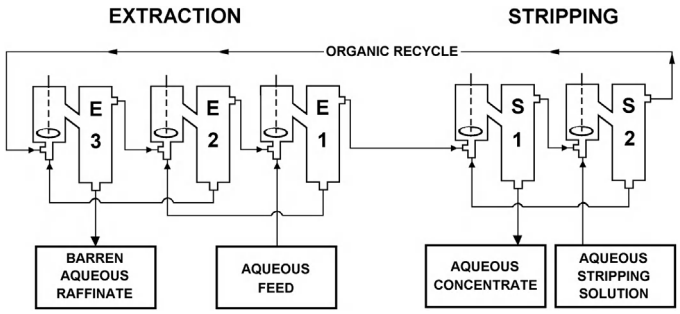


Figure A.2 Mixer-settler counter current solvent extraction battery.

batteries containing any number of stages. In these batteries the aqueous and organic phases flow counter current to each other (Fig. A.2).

For counter current solvent extraction, either batch or continuously, one finds that for the stationary state and for n stages

$$\varphi_n = (P-1)/\left(P^{n+1}-1\right) \tag{A.7}$$

provided P is constant through all stages. In our example one obtains for $\theta = 0.5$ (flow rate ratio organic: aqueous) and $n = 3$ that $E_U = 99.91\%$ and $E_{La} = 3.5\%$. Thus about the same yield of uranium, but with a somewhat lower lanthanum impurity as compared to the crosscurrent extraction procedure. This impurity figure can be lowered substantially by modification of the extraction process according to Fig. A.3 so that the extraction battery contains n extraction stages with extraction factor P_1, and $m-1$ *washing* stages with the *extraction* factor P_2. Then

$$\varphi_{m,n} = \frac{(P_1-1)\left(P_2^{m}-1\right)}{\left(P_1^{n+1}-1\right)(P_2-1)P_2^{m-1}+\left(P_2^{m-1}-1\right)(P_1-1)} \tag{A.8}$$

which in the case that $P_1 = P_2$ reduces to

$$\varphi_{m,n} = (P^m-1)/\left(P^{m+n}-1\right) \tag{A.9}$$

In the latter case, using three extraction stages and one washing stage, we find for our example that E_U is 99.9% and E_{La} is 0.12%. Thus both high yield and high purity are

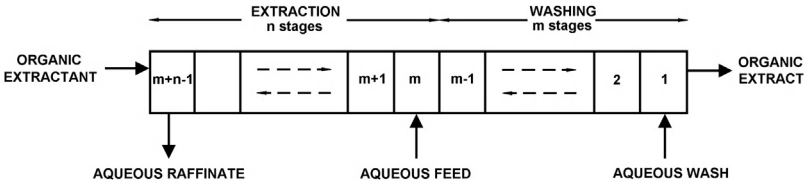


Figure A.3 Counter current solvent extraction with n extraction and m washing stages.

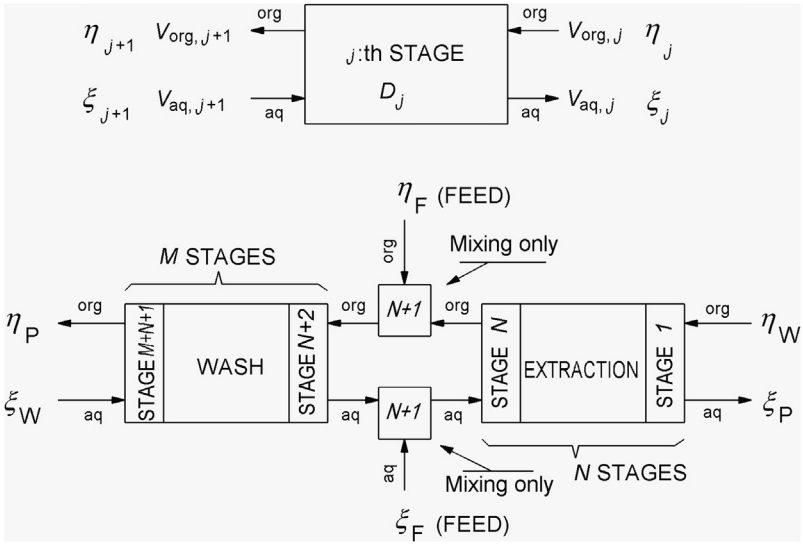


Figure A.4 Notations for counter current extraction with wash stages.

achieved with the counter current, central or intermediate feed solvent extraction technique. This is extensively used in uranium production, nuclear fuel reprocessing, and transuranium element separations. When the conditions are selected so that a high $(P \gg 1)$ extraction factor is obtained for the desired product and low ones $(P \ll 1)$ for the impurities, high purity and good yield can be obtained with relatively few stages. However, θ is always lower in the extraction stages than in the washing stages because the central feed flow rate adds to the wash flow rate in the extraction stages and the organic flow rate is unchanged. Hence, cases with the same P-value in extraction and washing stages seldom occur.

The purpose of the *washing* (or *scrubbing)* stages is to clean the desired product from impurities, and should not be confused with the *stripping* stages, the purpose of which is to transfer the product from the organic phase to a new aqueous phase. However, stripping can also be so arranged that further purification is achieved, sometimes even using a central organic feed, cf. Fig. A.4. If no additional purification is desired after extraction and the extractant is volatile, the organic phase may be distilled, leaving a pure solid product.

A.3. HIGH LOADINGS

A further industrial requirement on solvent extraction processes is high capacity, which means high concentrations of several solutes in the aqueous and organic solvents ('high loading'). The extraction factor P (or distribution factor D_M) may then vary from stage to